Amazon Sales Analysis Project

Using SQL Insights to Reduce Order Cancellations & Improve Revenue

Project Objective

- Analyze top cities with highest return & cancellation rates
- Compare order amounts, product categories, and customer behavior
- Recommend SQL-based strategies to reduce losses and improve satisfaction

Dataset Overview

- amazon_orders: Order status, amounts, dates
- amazon_customers: City, email, name
- amazon_products: Product categories
- amazon_payments: Payment methods, payment amount

Top Return & Cancellation Cities

- Top 5 cities have highest rate of returned/cancelled orders
- Delivery status used: 'Returned' or 'Cancelled'
- Ahmedabad, Mumbai, etc. appeared at the top

Product Category Insights

- Clothing category had highest return rate
- Return rate per category calculated
- Focus needed on top 3 return-prone categories

Customer Behavior Insights

- Repeat customers by email & name found
- Customers with >5 orders identified
- Potential loyalty program targets

Payment & Revenue Insights

- Most used: Credit Card and UPI
- Highest revenue via Credit Card
- Supports focused financial channel promotions

City-Wise Revenue Loss

- Losses due to returned/cancelled orders calculated
- Major losses from metro cities
- Reform return policies regionally

Recommendations

- Improve product quality in top-return categories
- Review policies in high-return cities
- Promote reliable payment methods
- Launch loyalty rewards for high-frequency customers

Conclusion

- Identified top loss-making cities and categories
- SQL-based insights guided problem-solving
- Proposed actionable business improvements